

ASAF BELILUS

M.Sc. Computer Science Researcher · Computer Vision & Biomechanics

CONTACT

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[LinkedIn](#)
[GitHub](#)
[Portfolio](#)

SKILLS & TECHNOLOGIES

Languages

Python, Java, C/C++, TypeScript

Research Tooling

NumPy, SciPy (nonlinear optimization),
kinematics/CV pipelines

Backend

Spring Boot, REST APIs, PostgreSQL

Tools & Testing

Docker, Git, automated regression testing

EDUCATION

M.Sc. Computer Science

Ben-Gurion University of the Negev
Researcher, Prof. Gera Weiss's Lab · 2026 – Present (Expected Oct 2027)
Current GPA 95.

B.Sc. Computer Science

Ben-Gurion University of the Negev · 2022 – 2025
Final GPA 84.2 · Active swimmer, BGU ASA competitive swim team.
Coursework: Computational Models, Probability & Statistics, Intro to Computational Learning, Algorithms

ACTIVITIES

Teaching Assistant — BGU · 2026 – Present

Supervises practicum coursework and grades/examines student papers.

Individual/Group Swimming Instructor — Wingate Institute · 2014 – Present

Teaches fundamentals and advanced technique to swimmers of all ages.

Volunteer Math & CS Tutor — BGU · 2026 – Present

Tutors first-year students in math and CS, focused on reservist students returning to their studies.

LANGUAGES

Hebrew — Native
English — Fluent

PROFILE

Computer Science M.Sc. student seeking a research student position in computer vision, applied optimization, or scientific software. Reconstructs swimmer joint motion from underwater 3-D pose data for hydrodynamic simulation using Python, kinematics, and nonlinear least squares. Uses 111 regression tests and a staged error budget to separate software defects from measurement limits.

RESEARCH

Swimmer Biomechanics Research — Prof. Gera Weiss's Lab

Ben-Gurion University of the Negev · in collaboration with Prof. Raziel Reimer, Faculty of Engineering · 2026 – Present

- Conducting cross-disciplinary research applying computer vision and control theory to competitive-swimming biomechanics, bridging the Computer Science and Engineering faculties.
- Built Python pipelines that reconstruct swimmer joint motion from 3-D underwater pose-estimation data, using forward/inverse kinematics and nonlinear least-squares optimization to drive a third-party hydrodynamic simulation engine.
- Diagnosed systematic model-fitting errors (degenerate optimization bounds, skeletal geometry mismatches) via quantitative probing against ground-truth motion-capture data, backed by automated regression testing.

FEATURED PROJECT

SwimEdge — Full-Stack Competitive Swimming Platform

Founder & Sole Engineer · 2025 – Present · [GitHub](#)

- Designed and built a full-stack platform for competition management and performance analytics, motivated by a gap identified firsthand as a competitive swimmer.
- Engineered a modular, service-oriented Spring Boot + PostgreSQL backend and a custom PDF/Excel parsing pipeline for Israeli Swimming Association competition data, backed by continuously verified automated test suites.

MILITARY SERVICE

Combat Commander, Paratrooper Brigade, IDF

2016 – 2019

- Trained and mentored newly recruited soldiers; led operations and managed company-level logistics.

Reservist, Paratrooper Brigade, IDF

2019 – Present

- Lead frontline teams under pressure, maintaining calm and accountability.